FINDINGS REPORT

Small Business Revenue Forecasting

Using Machine Learning to Predict Monthly Revenue and Surface Actionable Business Insights

Tools: SQL · Python · XGBoost · Excel · Power BI
Dataset: Sample Superstore (Kaggle) · 9,994 transactions · 2014–2017

May 2026

Executive Summary

This report presents the findings from an end-to-end revenue forecasting project built on the Sample Superstore dataset. The goal was to build a machine learning model that predicts monthly revenue up to 30 days ahead, while surfacing actionable business insights about product performance, discount strategy, and seasonal patterns.

$2.30M	$286K	12.5%	8.3%
Total Revenue	*Total Profit*	*Profit Margin*	*Model MAPE*

Key Conclusions

- **XGBoost outperformed the baseline** Linear Regression model across all metrics — MAPE improved from 12.4% to 8.3%, meaning predictions landed within 8% of actual monthly revenue.
- **Lag features drove the biggest accuracy gain** — prior month revenue as a predictor contributed more to model performance than any feature engineering or model selection decision.
- **Discounts above 20% consistently destroy profit** — despite generating revenue, orders with discounts above 20% produce negative or near-zero margins across all categories.
- **Q4 seasonality is strong and predictable** — November and December consistently outperform other months by 40–60%, enabling reliable inventory and staffing planning.
- **Furniture Tables is a loss-making sub-category** — despite $207K in revenue, Tables generated a net loss of $17,725 due to deep discounting practices.

1. Data Overview

The analysis used the publicly available Sample Superstore dataset, comprising 9,994 order line items across 4 years (January 2014 – December 2017). The dataset covers three product categories — Furniture, Office Supplies, and Technology — sold across four US regions.

Dataset Summary

Attribute	Detail
Source	Kaggle — Sample Superstore (Tableau)
Rows	9,994 order lines
Columns	21 features
Date Range	January 2014 – December 2017 (4 years)
Categories	Furniture, Office Supplies, Technology
Regions	West, East, Central, South
Missing Values	None — clean dataset
Target Variable	Monthly Sales (aggregated from order lines)

2. Methodology

Step 1: SQL Extraction	Raw CSV loaded into PostgreSQL. Dimension tables (customer, product) and a fact table created. Five summary views built for dashboard consumption.
Step 2: Exploratory Analysis	Monthly revenue trends, category breakdowns, and discount-vs-profit relationships visualised in Python (matplotlib/seaborn).
Step 3: Feature Engineering	Lag features (prior 1, 2, 3 months revenue), rolling 3-month and 6-month averages, and calendar features (month, quarter) created.
Step 4: Modelling	Linear Regression trained as baseline. XGBoost trained with 200 estimators, learning rate 0.05, max depth 4. Time-ordered train/test split used (last 6 months as test).
Step 5: Evaluation	MAE, RMSE, and MAPE calculated. Actual vs predicted chart and feature importance plot generated.
Step 6: Dashboards	Results surfaced in a 5-sheet Excel workbook and a 4-page Power BI dashboard.

3. Model Results

Two models were evaluated on the final 6 months of data (held out as a test set). XGBoost substantially outperformed the Linear Regression baseline on all metrics.

Model	MAE	RMSE	MAPE	Verdict
Linear Regression	$4,200	$5,100	12.4%	Baseline
XGBoost	$2,850	$3,400	8.3%	Selected

Interpretation

For a business turning over approximately $50,000–$80,000 per month (as in the test period), an 8.3% MAPE means predictions are typically within $4,000–$6,600 of actual revenue. This is **actionable accuracy** — sufficient for staffing, procurement, and cash flow planning decisions.

Top Predictive Features

XGBoost's feature importance scores revealed the following ranking:

Rank	Feature	Importance	Interpretation
1	lag_1 (prior month revenue)	High	Momentum carries forward month-to-month
2	rolling_3m (3-month avg)	High	Smoothed trend is highly predictive
3	Quarter	Med	Q4 seasonality captured
4	lag_2 (2 months prior)	Med	Secondary momentum signal
5	Month	Med	Intra-year seasonal patterns
6	lag_3 (3 months prior)	Low	Diminishing return at 3-month lag

4. Business Insights

4.1 Discount Strategy

One of the most significant findings of this analysis concerns the relationship between discount rates and profitability. The data shows a clear inflection point at the 20% discount threshold.

Discount Band	Orders	Revenue	Profit	Margin %
0% – No Discount	3,876	$956K	$198K	20.7%
1% – 10%	2,341	$587K	$89K	15.2%
11% – 20%	1,823	$424K	$31K	7.4%
21% – 30%	1,204	$231K	-$8K	-3.5%
31%+	750	$99K	-$24K	-24.7%

Recommendation	Cap all discounts at 20%. Orders above this threshold lose money on average. A discount ceiling policy could recover an estimated $32K+ in annual profit currently being given away through excessive discounting.

4.2 Category Performance

Category	Sub-Category	Revenue	Profit	Margin %	Action
Technology	Copiers	$149K	$55K	37.2%	Grow
Technology	Accessories	$167K	$42K	25.1%	Grow
Office Sup.	Paper	$78K	$34K	43.4%	Grow
Furniture	Chairs	$328K	$27K	8.1%	Maintain
Furniture	Bookcases	$115K	-$3K	-3.0%	Review
Furniture	Tables	$207K	-$18K	-8.6%	Discontinue/Fix

4.3 Seasonality

Revenue follows a consistent seasonal pattern across all four years:

- **Q1 (Jan–Mar):** Weakest quarter — revenue 25–35% below annual average.
- **Q2–Q3 (Apr–Sep):** Steady growth, approaching average.
- **Q4 (Oct–Dec):** Strongest quarter — November and December peak at 40–60% above average.

Recommendation	Use Q4 projections to pre-position inventory by October. Consider targeted Q1 promotions (max 15% discount) to smooth the seasonal dip without eroding margin.

5. Recommendations

High	**Implement a 20% discount ceiling** Immediately cap all promotional discounts at 20%. Orders above this rate are loss-making on average. Enforce via pricing policy or POS system rules.	*Finance / Sales*
High	**Prioritise Technology and Paper sub-categories** Copiers (37.2% margin) and Paper (43.4% margin) are the highest-margin sub-categories. Increase inventory allocation and marketing budget toward these lines.	*Procurement*
High	**Review Furniture Tables pricing** Tables generated $207K in revenue but a net loss of $17,725. Investigate whether deep discounting on Tables is driving this or if the cost structure is unviable.	*Category Management*
Medium	**Use Q4 forecast for inventory planning** The model predicts Q4 revenue with 8.3% MAPE. Use October forecasts to pre-order stock for November–December peak, reducing stockout risk.	*Operations*
Medium	**Deploy the model to a simple dashboard** Wrap the XGBoost model in a Streamlit app so non-technical staff can generate monthly forecasts by entering the prior month's revenue figure.	*Technology*

6. Limitations & Next Steps

Limitations: The model was trained on a single retail dataset and may not generalise directly to other business contexts. The 6-month test set is small; longer evaluation would strengthen confidence. External factors (economic conditions, competitor pricing) are not captured.

Next steps: (1) Retrain on fresh data quarterly. (2) Add external signals — web traffic, ad spend, promotions calendar. (3) Build a Streamlit forecasting app. (4) Extend to sub-category level forecasting for more granular planning.

This report was produced as part of a Data Science Portfolio project. All data is publicly available via Kaggle. | May 2026